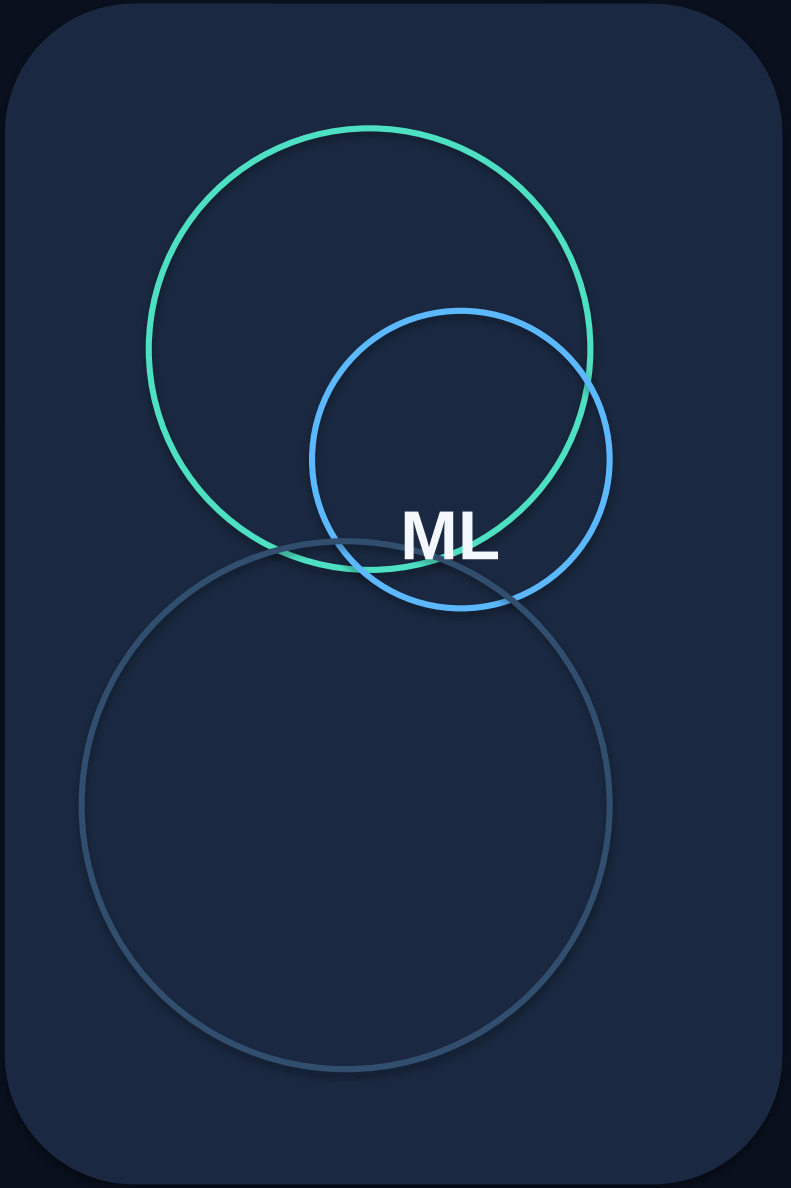
AI FRAUD SENTINEL

Machine learning to identify and prioritize suspicious transactions

2025 HISTORY → JAN 2026 SCORING

Built to support human review, not automatic customer blocking.



From manual review to risk-based prioritization

THE CHALLENGE

More transactions, limited manual capacity

- Fraud may go undetected
- Incorrect declines affect legitimate customers

THE APPROACH

A score to organize the queue

- Estimate the risk of each transaction
- Prioritize cases for human review

SOLUTION FLOW

01 Data → 02 Features → 03 CatBoost → 04 Score → 05 Review

A large dataset, with every record accounted for

523,525

Historical transactions

January to December 2025

44,451

Transactions to score

January 2026

3.91%

Labeled fraud cases

20,485 positive records

0

Missing values

No duplicate rows either

DATA STRUCTURE

2025 | 15 colunas

Includes `flag_fraude`, the target variable.

JAN/2026 | 14 colunas

No ground-truth label; records are scored by the model.

INTEGRITY

567,976

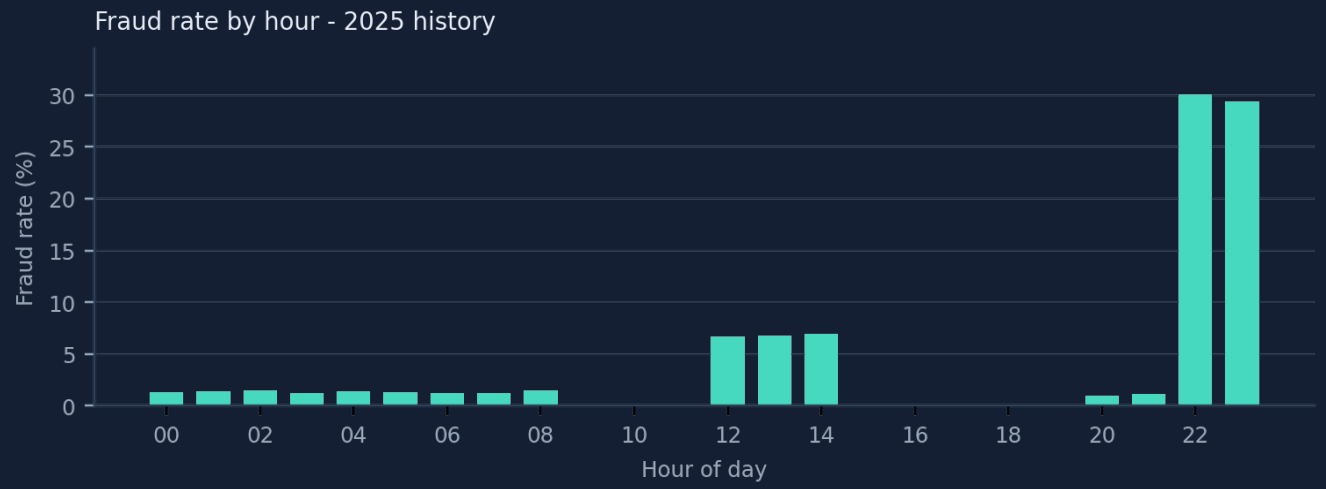
rows loaded across both datasets

All 14 original January columns are retained. The output CSV adds 3 prediction fields.

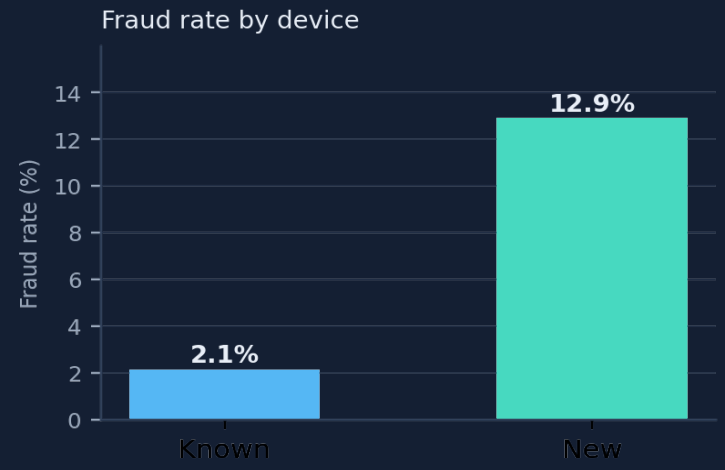
Important: January classifications are model predictions, not confirmed fraud; the source file has no observed outcome.

Signals vary by transaction hour and device

FRAUD RATE BY HOUR - 2025 HISTORY



NEW VS. KNOWN DEVICE



12.9% vs. 2.1% historical rate for new vs. known devices

49.5% of the model importance is attributed to transaction hour

Note: association is not causation. Hours with no labeled fraud should be investigated before they inform operational rules.

A time-based split to simulate future use

TIME-BASED SPLIT

JAN-SET/2025
Training
391,566

OUT-NOV/2025
Threshold calibration
87,508

DEZ/2025
Final test
44,451

No shuffling: the model learns from earlier records and is evaluated on a later month.

MODEL

CatBoost Classifier

Gradient boosting with native handling of categorical features.

0.293
initial threshold

74
best iteration

The threshold was selected by the best F1 on Oct-Nov. In production, it should reflect the cost of missed fraud and unnecessary alerts.

Performance on a month held out from training

0.910

ROC-AUC

Ranking ability

0.609

Average Precision

Baseline = 0.054

62.8%

Precision

Share of alerts that were fraud

57.6%

Recall

Fraud cases identified

CONFUSION MATRIX

41,218 True negative	**822** False positive
1,023 False negative	**1,388** True positive

2,210 alerts

4.97% of test-month volume

The model identified 1,388 of the 2,411 fraud-labeled December transactions. It generated 822 false positives and missed 1,023 fraud cases.

Accuracy alone is not enough: labeling every transaction as legitimate would reach 94.6% accuracy in December, but recall would be zero.

January 2026: a prioritized review queue

44,451

Transactions scored

100% of source records

2,122

Predicted alerts

4.78% of volume

1,025

High risk

Review first

1,097

Review

Subject to review capacity

ALERT INSIGHTS

53.4% of predicted alerts occurred between 10 p.m. and 11:59 p.m.

The Fuel category had an 11.6% predicted alert rate within its transactions. This is a review signal, not a confirmed fraud rate.

NEXT STEPS

- Review alerts with analysts and estimate the cost of errors
- Monitor drift and performance as new labels become available
- Recalibrate the threshold before any blocking action

Key limitation: January has no ground-truth labels. Model precision for this month can only be measured once outcomes are confirmed.